Round 11 | Changing the represented state

Focused research assessment / 24 September 2026 / GPT-6 Astra

Verdict: I cannot provide a justified scheme meeting all three requirements: no κ_3 tensor or source legs, total cost ≤125 u, and a credible path to the chain's accuracy. The single mechanism assessed below is a **nonlinear closure with memory of the consumed channels**. Its missing update is substantive; it is not a completed estimator.

State and budget

Carry $q_l = (m_l, C_l, Y_l, k4_l)$, where $Y_{ij} = \text{kappa3}(h_i, h_i, h_j)$ for the current post-ReLU activations h. Variance and d3 are the diagonals of C and Y. All 16 channel snapshots need about $2Ln^2$ numbers: **256 MiB in float64**, plus vectors. This stores channel history, not dense source legs.

The known linear update $m' = Wm$, $C' = WCW^T$ costs two square products per layer. Across 16 layers that is **16.32 u** at 0.51 u/product. The entire 125 u allowance permits only **15.32 square products per layer** before elementwise work, leaving about 13.32 after covariance transport. This is a budget envelope, **not a FLOP count for a finished closure**.

The missing update

One must supply an explicit nonlinear rule from the stored history and fixed weights to the next Y and k4. The exact third-order target is the (2,1) slice after applying the next weight matrix M to each of the post-ReLU tensor K's three legs. Writing this target as a memory kernel does not evaluate it. General recurrent closure literature motivates retaining memory [1], but supplies no rule or accuracy bound for this fixed-weight ReLU problem. I have not constructed or trained such a rule.

Why the visible channels alone do not close

Let a, b, c be distinct. Add a symmetric perturbation H with value h at the six permutations of (a,b,c), zero elsewhere. Its current (2,1) slice and diagonal vanish. Nevertheless, with u = M[:,a], v = M[:,b], w = M[:,c], the next slice changes by

```
DeltaY = 2h [outer(u*v,w) + outer(u*w,v) + outer(v*w,u)],
Delta_d3 = 6h (u*v*w). Here * is entrywise multiplication.
```

This is a realized odd contribution under M → −M. Marginal and pairwise observables do not identify it. Extra nonlinear history might encode it; neither this example nor linear irreducibility excludes that possibility. Likewise, irreducibility is **not a proof of a universal 249 u arithmetic lower bound**.

One stored-capture falsifier - limited, explicit scope

At the L14 capture, inject H only into the hidden post-ReLU third-order tensor; keep all visible channel snapshots unchanged. Use the actual L14→L15 map M and compute ΔY above in $O(n^2)$, without forming H. Choose h so $\|\Delta Y\|_F = 0.06\, \|Y_{15}\|_F$. A rule seeing only the unchanged channel history returns the same answer in both cases and cannot attain 2% relative error in both. This falsifies a claim of **universal channel-state sufficiency**; it does **not** establish failure on the realizable random-MLP family. This injection test is specified, not executed here.

Research status: no qualifying ≤125 u mechanism established. The unresolved step is a cheap, deterministic, history-dependent rule for the hidden contribution and fourth-order channels. Giving it an unexplained fitted coefficient or a nominal $O(n^3)$ label would not answer Round 11.

[1] Ma, Wang & E (2018), Model Reduction with Memory and the Machine Learning of Dynamical Systems. Supports the memory-closure framework only; no ARC performance claim.

Local evidence: consultations/prompts/ROUND11_prompt_change_what_is_represented.md; writeup/WRITEUP_DRAFT_v3.md, §11. Project measurements are supplied evidence, not rerun results.